

CS 534 – Individual Assignment 3

End-to-End To-Do List (with latest data-quality notes)

Deliverable recap

One ZIP containing:

1. MachineFailurePredictorPipeline.py (entry-point `main()`)
2. **Raw CSV only** (`ai4i2020.csv`)
3. A PDF holding **Table 1 (CV results)** and **Table 2 (test-set results)**

0 Convert & Inspect Raw Data

✓	Task	Details
☐	0.1 Convert workbook → CSV	Save <code>ai4i2020.xlsx</code> as <code>ai4i2020.csv</code> ; commit the CSV to the repo (not the XLSX).
☐	0.2 Sanity check dataset	Print to console: • shape → 10 000 × 14 • zero missing values • Machine failure positives = 339 (3.39 %)
☐	0.3 Log correlations & red flags	High collinearity: <code>Air</code> ↔ <code>Process T</code> (+0.88) and <code>Torque</code> ↔ <code>Speed</code> (-0.88). Note in code comments.

1 Project Skeleton

✓	Task	Details
☐	1.1 Create repo & venv	Use PyCharm (or VS Code) with a clean virtual-env.
☐	1.2 Stub pipeline file	Functions: <code>load_data()</code> , <code>preprocess()</code> , <code>balance()</code> , <code>split_data()</code> , <code>tune_models()</code> , <code>evaluate()</code> , <code>main()</code> .

🔧 2 Load & Initial Printout

✓	Task	Console requirements (per rubric)
☐	2.1 Read CSV	Print first 5 rows of the seven kept columns.
☐	2.2 Describe data	Print <code>.info()</code> and class distribution.

🔧 3 Feature Engineering & Pre-processing

✓	Task	Details
☐	3.1 Drop IDs	Remove <code>UDI</code> and <code>Product ID</code> .
☐	3.2 One-hot encode Type	3 dummies → <code>Type_L</code> , <code>Type_M</code> , <code>Type_H</code> .
☐	3.3 Optional engineered features	• $\Delta T = \text{ProcessT} - \text{AirT}$ (HDF rule) • $\text{Power} = \text{Torque} \times \text{RotSpeed}$ (PWF rule) Then decide whether to drop one of the raw temps to curb multicollinearity.
☐	3.4 Scale numeric cols	Use <code>StandardScaler</code> inside a <code>ColumnTransformer</code> .
☐	3.5 Print transformed df shape	Required console output.

⚖️ 4 Class Re-balancing

✓	Task	Details
☐	4.1 Random under-sample majority	Goal: 339 failure + 339 normal = 678 rows.
☐	4.2 Stratify by Machine failure (+ Type)	Ensures all Type tiers survive in folds.
☐	4.3 Print new class counts	Confirm balance.

5 Train/Test Split

✓	Task	Details
☐	5.1 80 / 20 split	After balancing; use <code>test_size=0.2</code> , <code>stratify=y</code> .
☐	5.2 Hold test set unseen	No peeking until Step 7.

6 Model Tuning (5-fold CV, metric = MCC) — 25 pts

✓	Model	Hyper-parameter grid / notes
☐	Logistic Regression	<code>C</code> , <code>penalty</code> (<code>l1</code> , <code>l2</code>)
☐	SVM (RBF)	<code>C</code> , <code>gamma</code>
☐	KNN	<code>n_neighbors</code> , <code>weights</code> , <code>metric</code>
☐	Decision Tree	<code>max_depth</code> , <code>min_samples_split</code> , <code>class_weight</code>
☐	MLPClassifier	<code>hidden_layer_sizes</code> , <code>alpha</code> , <code>learning_rate_init</code>
☐	6.x Record CV results	Build Table 1 (model, best params, mean MCC ± std). Print to console.

7 Final Evaluation on Test Set

✓	Task	Details
☐	7.1 Refit each model on full train	Using its CV-best hyper-parameters.
☐	7.2 Compute test metrics	MCC (primary), Accuracy, Precision, Recall, F1.
☐	7.3 Populate Table 2	Print and save to PDF.
☐	7.4 Select winner	Highest test MCC; note tie-break rule in report.

8 Console & PDF Outputs

✓	Required printouts	Where
☐	Raw data head & info (Step 2)	Console
☐	Balanced class counts (Step 4)	Console
☐	Table 1 – CV scores	Console & PDF
☐	Table 2 – Test scores	Console & PDF

9 Package & Submit

✓	Task	Details
☐	9.1 Generate PDF	<code>matplotlib.table</code> , <code>pandas.to_latex</code> → <code>pdflatex</code> , or similar.
☐	9.2 Zip	<code>ai4i2020.csv</code> + <code>MachineFailurePredictorPipeline.py</code> + PDF.
☐	9.3 Verify reproducibility	<code>python MachineFailurePredictorPipeline.py</code> must reproduce all console prints and regenerate the PDF.

10 Polish & Document

✓	Task	Details
☐	10.1 PEP-8 & doc-strings	Each function with type hints, concise description.
☐	10.2 Logging	Use logging module (INFO level) for dataset stats and model progress.
☐	10.3 README (optional)	Short run instructions (helps grader).

★ 11 (Extra Credit) Chase Bonus

✓	Task	Details
☐	11.1 Alternative resampling or ensembling	E.g., SMOTE + bagging; submit best MCC.
☐	11.2 Aim top-2 MCC	Grants +10 pts per rubric.

Data-Quality Notes Recap

1. **No missing values** — skip imputation.
2. **Strong collinearity** — consider engineered features (ΔT , Power) or drop redundant originals.
3. **Severe class imbalance** — under-sample majority class to 1:1 ratio before split.
4. **Minor Type_H sparsity** — maintain its presence via stratification.

Include these checks in pipeline logging so graders see you validated the dataset.